DSSSB • TGT COMPUTER SCIENCE • POST 41/26

Part A
Question-type workbook

Arithmetic, reasoning, English, Hindi and general awareness. Questions first, with two additional practice questions in every regular section.

30 days | 3–4 hours daily | English explanations

257 worked questions • Prepared 20 October 2026

How to use: Attempt every question in a type before reading its solutions. Cover the answers on paper, or use the answer toggle in the HTML workbook. Redo errors after 1, 3 and 7 days.

Coverage: One section is provided for each mapped syllabus/question family. Each section places practice questions first, then worked answers and a shortcut or solving rule. Each regular Part A/B section now includes two new questions marked Additional practice. Original pattern-based expected practice is included throughout; it is informed practice, not a prediction or a claim to reproduce every historical question.

Syllabus status: The 2026 advertisement verifies the exam scheme for 41/26. The supplied 56/24 subject syllabus is used provisionally: a separate 41/26 subject-syllabus PDF was not verified. Follow any later official 41/26 notice. Exact exam date was not supplied; timetable days are relative.

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A01 • ORIGINAL PRACTICE

Number system, divisibility and remainders

Observed: P35 · TGT Computer Science 2021 file 3 (2021) Q44; P33 · TGT Computer Science 2021 file 1 (2021) Q41-45

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A01.1 What smallest digit x makes 43x2 divisible by 9?

A01.2 Find the remainder when 7^103 is divided by 6.

A01.3 How many integers from 1 to 100 inclusive are divisible by 6 or 8?

A01.4 Additional practice: Find the least number to add to 999 so it is divisible by 16.

A01.5 Additional practice: What is the units digit of 7^2026?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Use digit sums for 3/9; last two digits for 4; last three for 8; alternating digit-sum difference for 11. Reduce each factor modulo the divisor. Divisibility by 6 requires BOTH 2 and 3.

A01.1 0. The digit sum is 9+x; x=0 or 9, and the smallest is 0.

A01.2 1. Since 7 leaves remainder 1, every positive power also leaves 1.

A01.3 24. floor(100/6)+floor(100/8)-floor(100/24)=16+12-4. Subtract the overlap once.

A01.4 9. 999=16×62+7; the next multiple needs 16−7=9. For “add” use divisor minus remainder; if already divisible, add zero unless a positive addition is required.

A01.5 9. Powers of 7 cycle through 7,9,3,1. Since 2026 leaves remainder 2 on division by 4, use the second entry.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A02 • ORIGINAL PRACTICE

HCF, LCM and bells

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q45; P35 · TGT Computer Science 2021 file 3 (2021) Q45

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A02.1 Find HCF and LCM of 36 and 60.

A02.2 Two numbers have HCF 12 and LCM 180. If one is 36, find the other.

A02.3 Bells ring every 12, 15 and 20 minutes, together at 9:00. When next together?

A02.4 Additional practice: Three lights flash every 18,24 and 30 seconds. How many times do all flash together in the next 30 minutes, excluding the initial flash?

A02.5 Additional practice: What is the largest number dividing 85 and 121 with remainders 1 and 1 respectively?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

For two positive integers only: product = HCF × LCM. Use prime factors. Repeated simultaneous events use LCM; largest equal groups use HCF.

A02.1 HCF=12; LCM=180. Common minimum powers give 12; 36×60/12=180.

A02.2 60. 12×180/36.

A02.3 10:00. LCM=60 minutes. Do not add the intervals.

A02.4 5. LCM=360 seconds; 1800/360=5. Including the initial flash would give 6.

A02.5 12. Compute HCF(85−1,121−1)=HCF(84,120)=12. Subtract the specified remainders first.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A03 • ORIGINAL PRACTICE

Simplification, decimals and fractions

Observed: P35 · TGT Computer Science 2021 file 3 (2021) Q41-43

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A03.1 Evaluate 144 ÷ 12 × 3 + 7.

A03.2 Evaluate 0.0048 ÷ 0.12.

A03.3 Which is larger, 7/12 or 5/9?

A03.4 Additional practice: Evaluate 2¾ + 1⅚ − ⅔.

A03.5 Additional practice: Evaluate √0.0081 + (0.2)^3.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Do brackets and powers first, then multiplication/division left to right, then addition/subtraction left to right. Cross-multiply positive fractions. Approximate only when the question permits it.

A03.1 43. 12×3+7. Multiplication does not automatically precede division.

A03.2 0.04. Multiply numerator and denominator by 100: 0.48/12.

A03.3 7/12. Compare 7×9=63 with 5×12=60; no decimal division needed.

A03.4 47/12=3 11/12. Convert to improper fractions and use denominator 12: 33/12+22/12−8/12.

A03.5 0.098. √0.0081=0.09 and 0.2 cubed=0.008. Keep powers separate before adding.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A04 • ORIGINAL PRACTICE

Percentages: direct, reverse and successive

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q47; P35 · TGT Computer Science 2021 file 3 (2021) Q47-48

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A04.1 Find 12.5% of 640.

A04.2 A price increases 20%, then falls 20%. Find the net change.

A04.3 A is 25% greater than B. B is what percent less than A?

A04.4 Additional practice: A price rises by 25%. By what percentage must consumption fall to keep expenditure unchanged?

A04.5 Additional practice: Passing requires 40% marks. A student scores 144 and fails by 16. What are the maximum marks?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Convert familiar percentages to fractions: 12.5%=1/8, 16⅔%=1/6, 20%=1/5, 25%=1/4. Successive changes multiply factors; they do not simply add.

A04.1 80. Divide by 8.

A04.2 4% decrease. 1.20×0.80=0.96; equal opposite changes do not cancel.

A04.3 20%. Take B=100, A=125; difference/base=25/125.

A04.4 20%. New consumption factor=1/1.25=0.8. Shortcut: reduction=100r/(100+r) for a price rise r%.

A04.5 400. Passing marks=144+16=160; 160/0.40=400. Recover the base before applying the percentage.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A05 • ORIGINAL PRACTICE

Ratios, proportion and mixtures

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q48,50; P20 · TGT SOCIAL SCIENCE FEMALE 15th Oct 2018 Shift 1 (2018) Q51 for mixture

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A05.1 A:B=2:3 and B:C=4:5. Find A:B:C.

A05.2 Two numbers have sum 84 and difference 36. Find their ratio.

A05.3 Mix rice costing Rs 40/kg and Rs 70/kg to obtain Rs 50/kg. Find cheap:dear quantity ratio.

A05.4 Additional practice: A 20-litre mixture has milk:water=3:1. How much water makes the ratio 3:2?

A05.5 Additional practice: Mix rice at ₹30/kg and ₹42/kg to obtain ₹34/kg. Find the quantity ratio of cheap to dear rice.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Scale ratios to a common term. If (large-small)/(large+small)=k, large:small=(1+k):(1-k). Alligation uses cross-differences only when quantities blend linearly.

A05.1 8:12:15. Multiply the first ratio by 4 and the second by 3.

A05.2 5:2. Numbers=(84±36)/2=60,24.

A05.3 2:1. (70-50):(50-40)=20:10.

A05.4 5 litres. Milk=15 and water=5; required water=10, so add 5. The milk quantity remains fixed.

A05.5 2:1. Alligation gives (42−34):(34−30)=8:4. Use the mean between the two prices; no profit is included here.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A06 • ORIGINAL PRACTICE

Averages and ages

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q46; P09 · TGT MATHS FEMALE 22nd September 2018 Shift 1 (2018) Q16

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A06.1 The average of 8 values is 15. Adding one value makes the average 17. Find it.

A06.2 Replacing a value 20 increases the average of 10 values by 3. Find the replacement.

A06.3 A is twice B now. Six years later their ages are in ratio 3:2. Find present ages.

A06.4 Additional practice: The average of 12 observations is 18. One value, 32, was recorded as 23. Find the corrected average.

A06.5 Additional practice: A father is three times his son's age. After 12 years he will be twice as old. Find their present ages.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Total = average × count. For replacement, total change = count × average change. Age differences remain constant; ratios do not.

A06.1 33. New total 9×17=153 minus old total 8×15=120.

A06.2 50. Total increases by 30; new value=20+30.

A06.3 A=12, B=6. Let B=x: (2x+6)/(x+6)=3/2 gives x=6.

A06.4 18.75. Correct total=12×18+(32−23)=225; divide by 12. Correct the total rather than recomputing all values.

A06.5 Son 12, father 36. Put ages x and 3x: 3x+12=2(x+12), so x=12. Their age difference stays constant.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A07 • ORIGINAL PRACTICE

Profit, loss and marked-price discount

Observed: P35 · TGT Computer Science 2021 file 3 (2021) Q46,49; P09 · TGT MATHS FEMALE 22nd September 2018 Shift 1 (2018) Q49

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A07.1 An article sells for Rs 960 at 20% profit. Find cost price.

A07.2 Marked price is Rs 2,000. Two discounts of 10% and 20% apply. Find selling price and equivalent discount.

A07.3 Two articles are each sold for Rs 1,200, one at 20% gain and one at 20% loss. Find overall result.

A07.4 Additional practice: A shopkeeper marks an item 40% above cost and allows a 20% discount. Find the profit percentage.

A07.5 Additional practice: A trader uses a 900 g weight while charging the cost price of 1 kg. Find the gain percentage.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Profit/loss percent uses COST price; discount uses MARKED price. SP=CP×(1±rate). Work backwards by dividing by the multiplier.

A07.1 Rs 800. 960/1.20.

A07.2 Rs 1,440; 28%. 2000×0.9×0.8. Equivalent discount=10+20-(10×20/100).

A07.3 Rs 100 loss, or 4%. Costs are 1000 and 1500; total SP=2400, CP=2500. Equal SP and equal opposite percentages imply loss=r^2/100 percent.

A07.4 12%. Selling-price factor=1.4×0.8=1.12. Take cost as 100 to avoid unnecessary algebra.

A07.5 11⅑%. Receive the price of 1000 g for cost of 900 g. Gain=100/900×100. The denominator is actual cost, not the claimed weight.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A08 • ORIGINAL PRACTICE

Simple and compound interest

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q52; P20 · TGT SOCIAL SCIENCE FEMALE 15th Oct 2018 Shift 1 (2018) Q52

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A08.1 Find SI on Rs 5,000 at 8% yearly for 3 years.

A08.2 Find CI on Rs 10,000 at 10% annually for 2 years.

A08.3 Find the amount on Rs 8,000 at 10% yearly compounded half-yearly for one year.

A08.4 Additional practice: A sum doubles in 8 years at simple interest. In how many years does it triple at the same rate?

A08.5 Additional practice: Find the difference between compound and simple interest on ₹5,000 at 10% annually for 2 years.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

SI=Prt/100. Annual compound amount=P(1+r/100)^t. Half-yearly: halve rate and double periods. For two years, CI-SI=P(r/100)².

A08.1 Rs 1,200. 5000×8×3/100.

A08.2 Rs 2,100. Amount=10000×1.1²=12100.

A08.3 Rs 8,820. 8000×1.05²; interest=820.

A08.4 16 years. Doubling earns one principal in 8 years; tripling requires interest of two principals. This proportional shortcut is for simple interest.

A08.5 ₹50. For two annual periods the difference is P(r/100)^2=5000×0.01. Do not use this two-year shortcut unchanged for three years.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A09 • ORIGINAL PRACTICE

Time-work, efficiency, wages and pipes

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q55; P35 · TGT Computer Science 2021 file 3 (2021) Q55; pipes added for coverage

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A09.1 A finishes in 12 days and B in 18. Find their joint completion time.

A09.2 An inlet fills a tank in 6 hours; an outlet empties it in 9. Time with both open?

A09.3 A does 3 units/hour for 4 hours; B does 2 units/hour for 6 hours. Split Rs 1,800.

A09.4 Additional practice: A fills a tank in 12 hours, B in 18 hours, and a drain empties it in 36 hours. How long together?

A09.5 Additional practice: A completes a job in 12 days and B in 18 days. Their total wage is ₹1,000. What is A's share for working together throughout?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Take total work as LCM of completion times. Add work rates; an outlet has a negative rate. Wages follow work done, not time alone.

A09.1 36/5=7.2 days. Total work 36; rates 3 and 2; 36/5.

A09.2 18 hours. Net rate=1/6-1/9=1/18.

A09.3 Rs 900 each. Both do 12 units. Do not divide wages 4:6.

A09.4 9 hours. Use 36 capacity units: rates +3,+2,−1, net 4; time 36/4. Assume all rates are constant and the drain operates throughout.

A09.5 ₹600. Efficiency ratio=(1/12):(1/18)=3:2. Divide wages in the ratio of work done, not the ratio of completion times.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A10 • ORIGINAL PRACTICE

Workforce changes

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q55

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A10.1 Ten workers finish a job in 12 days at 6 hours/day. How long for 15 workers at 8 hours/day?

A10.2 Eight workers can finish in 15 days. After five days, three leave. How many further days?

A10.3 A is 50% more efficient than B. If B needs 18 days, how long does A need?

A10.4 Additional practice: Twelve workers finish a job in 15 days. After 5 days, 3 workers leave. How many additional days are needed?

A10.5 Additional practice: Eight people working 6 hours daily complete a job in 10 days. How many people finish it in 8 days at 5 hours daily?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

With equal efficiency: workers × hours/day × days is proportional to completed work. Calculate remaining work before changing the workforce.

A10.1 6 days. Total=720 worker-hours; daily capacity=120.

A10.2 16. Remaining=8×10=80 worker-days; 80/5=16.

A10.3 12 days. Rate ratio=3:2, time ratio=2:3.

A10.4 13⅓ days. Total work=180 worker-days; completed=60, remaining=120. Nine workers need 120/9 days, assuming equal productivity.

A10.5 12. Work=8×6×10=480 person-hours; each new worker contributes 8×5=40 hours. Divide 480 by 40.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A11 • ORIGINAL PRACTICE

Speed, relative motion and average speed

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q53

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A11.1 A car covers 150 km in 2.5 hours. Find speed in m/s.

A11.2 A journey has equal distances at 40 and 60 km/h. Find average speed.

A11.3 Two runners move in the same direction at 8 and 6 m/s, initially 100 m apart. Catch-up time?

A11.4 Additional practice: A car covers equal distances at 40 and 60 km/h. Find average speed.

A11.5 Additional practice: Two cyclists start 90 km apart and approach each other at 12 and 18 km/h. When do they meet?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Convert km/h to m/s by ×5/18. Equal-distance average speed=2uv/(u+v); equal-time average=(u+v)/2. Relative speeds add for opposite directions.

A11.1 60 km/h=50/3 m/s.

A11.2 48 km/h. 2×40×60/(40+60).

A11.3 50 seconds. Relative speed=2 m/s; 100/2.

A11.4 48 km/h. For equal distances use 2uv/(u+v)=4800/100. Arithmetic averaging would apply to equal times instead.

A11.5 3 hours. Closing speed=12+18=30 km/h; time=90/30. Add speeds only because separation is shrinking from opposite directions.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A12 • ORIGINAL PRACTICE

Trains, boats and wheels

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q54; P35 · TGT Computer Science 2021 file 3 (2021) Q53-54

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A12.1 A 180 m train at 54 km/h crosses a 120 m platform. Find time.

A12.2 Downstream speed is 18 km/h and upstream 10 km/h. Find boat and stream speeds.

A12.3 A wheel has circumference 2 m. How many turns over 3 km?

A12.4 Additional practice: A 150 m train passes a pole in 10 seconds. How long to pass a 300 m platform at the same speed?

A12.5 Additional practice: A boat's speed in still water is 10 km/h and the stream speed is 2 km/h. Find the time for 24 km upstream and 24 km downstream.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Train crossing distance is its length plus the object length. Boat still-water speed= (downstream+upstream)/2. Revolutions=distance/circumference, using the same units.

A12.1 20 seconds. Speed=15 m/s; distance=300 m.

A12.2 Boat=14 km/h; stream=4 km/h.

A12.3 1,500. 3000/2; convert kilometres first.

A12.4 30 seconds. Speed=15 m/s; platform crossing requires train+platform length=450 m. Divide 450 by 15.

A12.5 5 hours. Upstream=8, downstream=12 km/h; times=24/8+24/12=3+2. Add times, not the two speeds.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A13 • ORIGINAL PRACTICE

Plane mensuration

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q56; P35 · TGT Computer Science 2021 file 3 (2021) Q56

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A13.1 A rectangle has perimeter 50 cm and length 5 cm more than width. Find area.

A13.2 Find the area of an isosceles right triangle with hypotenuse 20 cm.

A13.3 A circle radius increases by 10%. Find percentage increase in area.

A13.4 Additional practice: A rectangle's length and breadth rise by 20% and 10%. What is the percentage increase in area?

A13.5 Additional practice: A circle and square have equal perimeter. Which has the larger area?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Rectangle area=lb; triangle=bh/2; circle=πr^2. Equilateral triangle area=$\sqrt{3}\,a^2/4$. Similar figures with length ratio k have area ratio k^2.

A13.1 150 cm^2. l+b=25, l-b=5 gives l=15,b=10.

A13.2 100 cm^2. Area=hypotenuse^2/4.

A13.3 21%. Area factor=1.1^2=1.21.

A13.4 32%. Multiply factors: 1.2×1.1=1.32. Equivalently a+b+ab/100 for two percentage increases.

A13.5 The circle. For perimeter P, areas are $P^2/(4\pi)$ and $P^2/16$; their ratio is $4/\pi>1$. Compare expressions using the common perimeter.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A14 • ORIGINAL PRACTICE

Solid mensuration and recasting

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q57; P35 · TGT Computer Science 2021 file 3 (2021) Q57

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A14.1 Find volume of a cylinder with r=7 cm,h=10 cm, using π=22/7.

A14.2 A solid cube of side 6 cm is cut into cubes of side 2 cm. How many?

A14.3 A sphere of radius 3 cm is melted into cones of radius 1 cm,height 3 cm. How many cones?

A14.4 Additional practice: How many spheres of radius 1 cm can be formed from one sphere of radius 3 cm, with no loss?

A14.5 Additional practice: A cylinder has radius 3 cm and height 7 cm. Find its curved surface area using π=22/7.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Cylinder $V=\pi r^2h$; cone $V=\pi r^2h/3$; sphere $V=4\pi r^3/3$. Hemisphere total surface=$3\pi r^2$, curved surface=$2\pi r^2$. Recasting conserves volume.

A14.1 1540 cm^3. π×49×10.

A14.2 27. $(6/2)^3$.

A14.3 36. Sphere volume=36π; each cone=π.

A14.4 27. Sphere volume is proportional to r^3; $(3/1)^3=27$. Surface areas do not remain conserved during melting.

A14.5 132 cm^2. Curved area=2πrh. Total surface area would also include two circular ends.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A15 • ORIGINAL PRACTICE

Data interpretation: table, line, bar and pie

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q58-60; P35 · TGT Computer Science 2021 file 3 (2021) Q58-60

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A15.1 Sales in Jan, Feb, Mar are 40, 50, 70 units. Find total and average.

A15.2 A line chart shows sales rising from 80 to 100. Find percentage increase.

A15.3 A pie chart represents Rs 72,000 spending; rent occupies 120°. Find rent.

A15.4 Revenue is Rs 150 lakh and expenditure Rs 120 lakh. Find profit as a percentage of expenditure.

A15.5 Additional practice: A class has 120 students: 45 walk, 30 cycle and the rest use buses. What is the bus sector angle in a pie chart?

A15.6 Additional practice: Sales were 80,100,120 and 100 units across four quarters. What percentage of annual sales occurred in the third quarter?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Read units and denominator before calculating. Percentage change uses the OLD value. A pie slice of θ degrees represents θ/360 of the total. A chart alone does not prove causation.

A15.1 Total=160; average=160/3=53⅓ units.

A15.2 25%. Increase 20 divided by old 80.

A15.3 Rs 24,000. 120/360=1/3.

A15.4 25%. Profit=30; 30/120×100. Dividing by revenue would give margin=20%, a different measure.

A15.5 135°. Bus users=45; fraction=45/120=3/8. Multiply by 360°, not by 100.

A15.6 30%. Total=400; third-quarter share=120/400×100. The base is the annual total, not the preceding quarter.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A16 • ORIGINAL PRACTICE

Number, alphabet and mixed series

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q21-22,32; P35 · TGT Computer Science 2021 file 3 (2021) Q21-22

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A16.1 Find the next term: 4, 9, 19, 39, ?.

A16.2 Find the next term: 2, 6, 12, 20, 30, ?.

A16.3 Complete A2, C6, E12, G20, ?.

A16.4 Additional practice: Find the next term: 3,8,15,24,35,?

A16.5 Additional practice: Complete: AZ, CX, EV, GT, ?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Try differences, second differences, multiplication±constant and alternating subsequences. For letters use A=1,...,Z=26 and wrap around. Check the rule against every given term.

A16.1 79. Multiply by 2 and add 1.

A16.2 42. Terms n(n+1), or differences 4,6,8,10,12.

A16.3 I30. Letters advance by 2; numbers follow n(n+1).

A16.4 48. Differences are 5,7,9,11, so add 13. Equivalent rule: n(n+2), beginning at n=1.

A16.5 IR. The first letters advance by 2; the second letters retreat by 2. Track columns separately.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A17 • ORIGINAL PRACTICE

Analogy and classification

Observed: P02 · TGT HINDI FEMALE 28th Aug 2018 Shift1 (2018) Q1; P13 · TGT NATURAL SCIENCE FEMALE 27th September 2018 Shift 1 (2018) Q1; P33 · TGT Computer Science 2021 file 1 (2021) Q23-25

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A17.1 Thermometer:temperature :: barometer:?.

A17.2 Which differs: 8,27,64,100?

A17.3 ACE:BDF :: MOQ:?.

A17.4 Additional practice: Book:author :: painting:?

A17.5 Additional practice: Which is the odd one: 16,25,36,49,63?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

State the relationship precisely: tool-use, animal-home, quantity-unit, part-whole or a numerical transformation. Odd-one-out requires one rule shared by the other three.

A17.1 Atmospheric pressure. Both are instrument-measured quantity pairs.

A17.2 100. The others are perfect cubes.

A17.3 NPR. Shift each letter forward one position.

A17.4 Painter. Preserve the object-to-creator relationship rather than choosing a place where paintings are displayed.

A17.5 63. The others are perfect squares. Prefer a simple property shared by every remaining member.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A18 • ORIGINAL PRACTICE

Coding-decoding

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q26; P35 · TGT Computer Science 2021 file 3 (2021) Q26

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A18.1 If CAT becomes DBU, encode DOG.

A18.2 If A=26,B=25,...,Z=1, find the code sum of CAB.

A18.3 “red big”=“ka mi”; “big box”=“mi tu”. What is the code for red?

A18.4 Additional practice: If CAT is coded as DBU, how is DOG coded by the same rule?

A18.5 Additional practice: In a code, “red big” is “ka mi” and “big box” is “mi tu”. What denotes “big”?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Test positional shifts, reversed alphabets (position 27-x), rearrangement and common-word intersections. Do not infer unique codes when the clues leave alternatives.

A18.1 EPH. Each letter moves +1.

A18.2 75. C=24,A=26,B=25.

A18.3 ka. Shared big=mi, leaving red=ka.

A18.4 EPH. Shift every letter forward by one. Verify the same transformation on all letters of the example.

A18.5 mi. Match the common word with the common code; the order of codes need not match the order of words.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A19 • ORIGINAL PRACTICE

Blood relations and family trees

Observed: P35 · TGT Computer Science 2021 file 3 (2021) Q27; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q9

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A19.1 A man says, “She is the daughter of my only son.” How is she related to him?

A19.2 P is Q’s sister; Q is R’s father. How is P related to R?

A19.3 A is B’s parent; B is C’s parent. Can A’s gender be determined?

A19.4 Additional practice: Pointing to a boy, Neha says, “His mother is the only daughter of my mother.” Who is the boy to Neha?

A19.5 Additional practice: A is B's brother; B is C's mother. What is A to C?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Draw one generation per horizontal level. Mark gender only when stated. “My father’s only son” depends on who is speaking; avoid assuming the speaker is male.

A19.1 Granddaughter. Son → his daughter.

A19.2 Paternal aunt. The relation is through R’s father.

A19.3 No. A is a grandparent; neither “grandfather” nor “grandmother” follows.

A19.4 Her son. The only daughter of Neha's mother is Neha herself, since Neha is female. Trace the relationship inside out.

A19.5 Maternal uncle. First record the generation and sex explicitly; being a sibling of the mother fixes the maternal relationship.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A20 • ORIGINAL PRACTICE

Directions, shadows and turns

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q36-37; P35 · TGT Computer Science 2021 file 3 (2021) Q36-37

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A20.1 Walk 6 m east,8 m north,6 m west. How far from start?

A20.2 Facing north, turn 90° clockwise then 180° anticlockwise. Final direction?

A20.3 At sunset a person's shadow lies to their right. Which way are they facing?

A20.4 Additional practice: Walk 6 km north, 8 km east and 6 km south. How far and in which direction are you from the start?

A20.5 Additional practice: A person facing west turns 90° clockwise and then 180° anticlockwise. Which direction now?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Use east=+x,north=+y. Net displacement uses coordinates, not total distance. Morning shadows point west; evening shadows east. Reduce rotations modulo 360°.

A20.1 8 m north. East-west motion cancels.

A20.2 West. N→E→W.

A20.3 North. Shadow is east, and east is to the right when facing north.

A20.4 8 km east. North and south cancel. Use net horizontal and vertical displacement, not total distance walked.

A20.5 South. West→north→south. Draw the compass if the order of turns causes confusion.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A21 • ORIGINAL PRACTICE

Ranking, order and alphabet positions

Observed: P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q2,4,6-7

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A21.1 A student is 12th from left and 19th from right. Total students?

A21.2 In a row of 40, Ravi is 15th from right. Meena is 7 places to his left. Her left rank?

A21.3 What is the 8th letter from the end of the alphabet?

A21.4 Additional practice: In a row of 42, Meera is 15th from the left. Find her rank from the right.

A21.5 Additional practice: How many English letters lie strictly between the 8th and 19th letters?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

When one person is counted from both ends: total=left rank+right rank-1. Opposite-end position in alphabet=27-position.

A21.1 30. 12+19-1.

A21.2 19th. Ravi=40-15+1=26; Meena=26-7.

A21.3 S. Position from start=27-8=19.

A21.4 28th. Right rank=N−left rank+1=42−15+1. The +1 accounts for the person being included from both ends.

A21.5 10. Difference−1=19−8−1. “Between” excludes both endpoints.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A22 • ORIGINAL PRACTICE

Linear, circular and floor arrangements

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q27,29; P35 · TGT Computer Science 2021 file 3 (2021) Q28-29; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q1

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A22.1 Five people face north. A is leftmost; E rightmost; B is immediately right of A; D immediately left of E. Who is in the middle?

A22.2 Four people face the centre. A is at the top, B immediately to A's left, C opposite A. Where is D?

A22.3 Four floors are numbered 1 lowest to 4 highest. P is above Q; Q on floor 2; R on floor 4. Where is P?

A22.4 Additional practice: A,B,C,D occupy four seats in a row. A is immediately left of B, C is immediately right of B, and D is at an end. Is the order uniquely determined?

A22.5 Additional practice: Six people sit around a circle facing the centre. If P is second clockwise from Q, where is Q relative to P?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Place the most constrained clue first. Fix one person in a circle to remove rotational duplicates. For inward-facing seats, left is clockwise; outward-facing reverses it. Keep separate cases when needed.

A22.1 C. Order A B C D E.

A22.2 Left-hand side of the page (west). B is on the page-right (east); C is bottom.

A22.3 Floor 3, leaving S on floor 1.

A22.4 No. Both ABCD and DABC satisfy the clues: A is immediately left of B, C immediately right of B, and D is at an end. Test all allowed placements before claiming uniqueness.

A22.5 Second anticlockwise. Reverse the direction while keeping the separation. Avoid translating clockwise into a person's left/right without checking their facing.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A23 • ORIGINAL PRACTICE

Syllogisms and Venn counting

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q30-31,34-35

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A23.1 All roses are flowers. No flower is metal. Does “No rose is metal” follow?

A23.2 Some cats are pets. All pets are animals. Does “Some cats are animals” follow?

A23.3 Of 60 pupils,35 study Hindi,30 English,15 both. How many study neither?

A23.4 Additional practice: All programmers are learners. No learner is idle. Does “No programmer is idle” follow?

A23.5 Additional practice: Of 60 students, 35 like tea and 30 like coffee; 10 like neither. How many like both?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Use only the premises. “Some A are B” converts to “Some B are A”; “All A are B” does not convert to “All B are A”. With universal premises alone, do not assume existence unless the exam states that convention.

A23.1 Yes. Rose is entirely inside flower, which does not overlap metal.

A23.2 Yes. The stated cat-pet overlap lies within animals.

A23.3 10. Union=35+30-15=50; neither=60-50.

A23.4 Yes. Programmers form a subset of learners and the learner set is disjoint from idle people. A Venn diagram proves the exclusion without assuming any programmers exist.

A23.5 15. At least one=60−10=50; intersection=35+30−50. Subtract “neither” before applying inclusion-exclusion.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A24 • ORIGINAL PRACTICE

Statements, assumptions, arguments and conclusions

Listed explicitly in Part A syllabus

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A24.1 “Use the footbridge to cross safely.” What essential assumption supports this advice?

A24.2 Should school drinking water be tested? Argument I: it can identify contamination. II: testing kits have attractive colours. Which is strong?

A24.3 All selected candidates passed a test. Neha passed it. Must she be selected?

A24.4 Additional practice: Statement: “Use the pedestrian bridge to cross safely.” Is it necessarily assumed that crossing on the road is always fatal?

A24.5 Additional practice: Statement: “Some buses are electric.” Conclusion: “All buses are electric.” Does it follow?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

An assumption is an unstated requirement, not just a related fact. Strong arguments address the decision with relevant consequences. A conclusion must follow; “may” does not mean “must”.

A24.1 The footbridge is available and usable for crossing. It does not require that all road crossings always cause accidents.

A24.2 I only. Safety evidence is relevant; colour is not.

A24.3 No. Passing is necessary here, not sufficient.

A24.4 No. A recommended safer option does not imply an absolute claim about every alternative. Reject assumptions stronger than the statement requires.

A24.5 No. An existential statement does not establish a universal one. Check the quantifier before applying real-world knowledge.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A25 • ORIGINAL PRACTICE

Cause-effect, assertion-reason and action

Listed explicitly in Part A syllabus

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A25.1 Ice-cream sales and heatstroke both rise in summer. Must ice cream cause heatstroke?

A25.2 Assertion: metal railway tracks expand on heating. Reason: thermal expansion increases their dimensions. Are both true and is the reason explanatory?

A25.3 A school notices repeated failed logins. Best first action: publish passwords, ignore it, or review logs and secure accounts?

A25.4 Additional practice: More umbrellas are sold on rainy days. Does this alone prove umbrellas cause rain?

A25.5 Additional practice: A school server contains student records and shows suspected unauthorised access. What is the best immediate action: publish records, ignore the alert, or isolate access and report through the incident procedure?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Correlation alone does not establish cause. A reason must explain the assertion, not merely be true. Prefer proportionate, lawful, practical action supported by the facts.

A25.1 No. Hot weather is a plausible common factor.

A25.2 Yes to both, and the reason explains the assertion.

A25.3 Review logs and secure accounts. Respond to evidence; do not expose credentials.

A25.4 No. Association does not establish that direction of causation. Rain can explain umbrella demand; the observation alone is insufficient for a causal proof.

A25.5 Isolate the affected access and report. Choose a proportionate action that limits harm and preserves evidence; avoid destroying records or making unsupported accusations.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A26 • ORIGINAL PRACTICE

Mathematical operators, eligibility and missing values

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q33; P35 · TGT Computer Science 2021 file 3 (2021) Q32-33; eligibility is syllabus extension

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A26.1 If @ means + and # means ×, evaluate 6 @ 3 # 4.

A26.2 Eligibility: age 21-30 inclusive and score at least 60. A candidate is 30 with score 60. Eligible?

A26.3 Rows are (2,3,13), (4,5,41), (6,7,?). Complete the last row using the same rule.

A26.4 Additional practice: Replace + with × and × with − in 8 + 3 × 5, then evaluate using normal precedence.

A26.5 Additional practice: Eligibility requires age at least 18 AND a certificate. Ravi is 19 but has no certificate. Is he eligible?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Replace coded operations first, then apply normal precedence. For eligibility check every condition, then any stated exception; do not invent an exception.

A26.1 18. 6+3×4.

A26.2 Yes. Both boundary values are included.

A26.3 85. Third value=sum of squares: 6^2+7^2.

A26.4 19. The transformed expression is 8×3−5=24−5. First substitute symbols, then apply precedence.

A26.5 No. An AND rule requires both conditions. Do not let one satisfied condition compensate for a missing mandatory one.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A27 • ORIGINAL PRACTICE

Calendars and clocks

Syllabus time-sequence extension

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A27.1 If today is Monday, what day is it after 45 days?

A27.2 Is 2100 a leap year?

A27.3 Find the smaller angle at 3:20.

A27.4 Additional practice: If 1 March is Monday in a non-leap year, what weekday is 1 April?

A27.5 Additional practice: Find the smaller angle between the hands at 4:20.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Ordinary year shifts the weekday by 1; leap year by 2 across a full year. Leap year: divisible by 4, except centuries unless divisible by 400. Clock angle=|30h-5.5m|; take smaller angle.

A27.1 Thursday. 45 mod 7=3.

A27.2 No. It is divisible by 100 but not 400.

A27.3 20°. Hour hand=100°, minute hand=120°. Do not hold the hour hand at 90°.

A27.4 Thursday. March contributes 31 days, or 3 odd days. Advance Monday by 3; leap-year status does not change March's length.

A27.5 10°. Hour hand=30×4+0.5×20=130°; minute hand=6×20=120°. Take the smaller angular difference.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A28 • ORIGINAL PRACTICE

Cubes, dice, mirrors and paper folding

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q38-40; P35 · TGT Computer Science 2021 file 3 (2021) Q38-40; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q13-15

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A28.1 A cube painted on all six faces is cut into 4×4×4 small cubes. How many have exactly two painted faces?

A28.2 For the same cube, how many small cubes are unpainted?

A28.3 A die shows face 1 adjacent to faces 2,3,4,5 across several views. Which face is opposite 1?

A28.4 A paper is folded once vertically. One hole is punched away from the fold. How many holes after opening?

A28.5 Additional practice: A cube is painted on all faces and cut into 4×4×4 equal cubes. How many have exactly two painted faces?

A28.6 Additional practice: What happens to the apparent left-right direction of an arrow pointing northeast when reflected in a vertical mirror?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Painted n×n×n cube, n≥2: three faces=8; two=12(n-2); one=6(n-2)²; none=(n-2)³. A vertical mirror reverses left-right. Unfold paper by reflecting across folds in reverse order.

A28.1 24. Twelve edges, each with 4-2 interior cubes.

A28.2 8. (4-2)³.

A28.3 6. A cube face has four adjacent faces and one opposite.

A28.4 Two, placed symmetrically across the fold. A hole exactly on the fold would be different.

A28.5 24. Each of 12 edges contributes n−2=2 non-corner cubes. Corners have three painted faces and must be excluded.

A28.6 It points northwest. A vertical mirror reverses the horizontal component while preserving the vertical component.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A29 • ORIGINAL PRACTICE

Figure reasoning and analytical judgment

Observed: P09 · TGT MATHS FEMALE 22nd September 2018 Shift 1 (2018) Q9-10; P13 · TGT NATURAL SCIENCE FEMALE 27th September 2018 Shift 1 (2018) Q3-5

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A29.1 An arrow points ↑,→,↓,← in successive figures. What comes next?

A29.2 A 2×2 grid contains how many squares of all sizes?

A29.3 You find a named identity card in a classroom. Best response?

A29.4 Additional practice: How many rectangles are in a grid of 3 rows and 4 columns, counting squares too?

A29.5 Additional practice: A paper is folded once along its vertical middle. A hole is punched away from the fold. How many holes appear on unfolding?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

For figure sequences track rotation, reflection, count, shading and position separately. Embedded figures may be rotated only if instructions permit. For situation reaction choose a useful response rather than a dramatic one.

A29.1 ↑. Rotate 90° clockwise each time.

A29.2 Five. Four small plus the full square.

A29.3 Hand it to the responsible school office for return. This is practical and preserves the owner’s privacy.

A29.4 60. Choose two of 4 horizontal boundary lines and two of 5 vertical lines: C(4,2)×C(5,2)=6×10.

A29.5 Two, at mirror-symmetric positions. A hole on the fold would behave differently, so check the punch's position.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A30 • ORIGINAL PRACTICE

English: agreement and tense

Observed: P35 · TGT Computer Science 2021 file 3 (2021) Q81-86

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A30.1 Correct: “Each of the students have a book.”

A30.2 Fill: “She has lived here ___ 2020.”

A30.3 Correct: “Neither the teacher nor the students is absent.”

A30.4 Additional practice: Correct the verb: “Each of the students have a notebook.”

A30.5 Additional practice: Fill the blank: “By the time we arrived, the film ___.”

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Find the real subject, ignoring intervening phrases. With neither...nor, standard exam usage agrees with the nearer subject. Since marks a starting point; for marks duration.

A30.1 “Each of the students has a book.” Each is singular.

A30.2 Since. 2020 is a starting point; “for six years” is a duration.

A30.3 “Neither the teacher nor the students are absent.” The nearer subject is plural.

A30.4 “Each of the students has a notebook.” The head subject “each” is singular; the plural phrase after “of” does not control agreement.

A30.5 “had started.” Use past perfect for an action completed before another stated past event.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A31 • ORIGINAL PRACTICE

English: articles, prepositions and comparison

Observed: P35 · TGT Computer Science 2021 file 3 (2021) Q81-84

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A31.1 Fill: “He is ___ honest man and ___ university teacher.”

A31.2 Correct: “She is more taller than her sister.”

A31.3 Fill: “She is interested ___ music and good ___ mathematics.”

A31.4 Additional practice: Choose a/an: “She is ___ honest officer.”

A31.5 Additional practice: Correct: “He is senior than me.”

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Choose a/an by sound. Avoid double comparatives/superlatives. Learn fixed prepositions in phrases rather than translating word for word.

A31.1 An; a. Honest begins with a vowel sound; university begins with a /y/ sound.

A31.2 “She is taller than her sister.” Do not use more with taller.

A31.3 In; at. These are standard collocations.

A31.4 “an.” The initial sound is a vowel because h is silent. Article choice follows pronunciation, not the first written letter.

A31.5 “He is senior to me.” In standard exam usage senior, junior, superior and inferior take “to”. Learn the adjective-preposition pair.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A32 • ORIGINAL PRACTICE

English: vocabulary, spelling and confusables

Observed: P35 · TGT Computer Science 2021 file 3 (2021) Q89-92

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A32.1 Give a synonym of “brief” and an antonym of “scarce”.

A32.2 Choose: accomodation / accommodation / acommodation.

A32.3 Fill: “The new rule will ___ attendance; its ___ will be reviewed.”

A32.4 Additional practice: Distinguish “principal” and “principle” in: “The ___ explained a scientific ___.”

A32.5 Additional practice: Choose the correctly spelt word: accommodate / accomodate / acommodate.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Use context to choose meaning and part of speech. Prefixes help but are not infallible. Keep a personal list of misspellings you actually make.

A32.1 Concise; abundant. Meaning must fit the sentence.

A32.2 Accommodation: double c and double m.

A32.3 Affect; effect. Here affect is a verb and effect a noun.

A32.4 “principal; principle.” Principal can mean the head of an institution; principle means a rule or underlying law. Use the grammatical and semantic context.

A32.5 Accommodate. It has double c and double m. Recall the whole word rather than guessing from pronunciation.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A33 • ORIGINAL PRACTICE

English: idioms and one-word substitution

Observed: P35 · TGT Computer Science 2021 file 3 (2021) Q93-95

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A33.1 What does “once in a blue moon” mean?

A33.2 A person who studies the origin and history of words is a/an ___.

A33.3 What does “spill the beans” mean?

A33.4 Additional practice: What does “spill the beans” mean?

A33.5 Additional practice: Give one word for “a person who can speak many languages”.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Treat idioms as units. Do not infer their literal meaning. Match the exact definition in one-word substitutions.

A33.1 Very rarely.

A33.2 Etymologist. An entomologist studies insects.

A33.3 Reveal a secret.

A33.4 Reveal a secret. Idioms usually require the established figurative meaning rather than a literal reading.

A33.5 Polyglot. Link poly- with “many”; do not confuse it with a linguist, whose work is the study of language.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A34 • ORIGINAL PRACTICE

English: voice, reported speech and sentence ordering

Observed: P35 · TGT Computer Science 2021 file 3 (2021) Q87-88; voice/reporting added for coverage

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A34.1 Convert to passive: “The teacher checked the notebooks.”

A34.2 Report later: Ravi said, “I am tired.”

A34.3 Order: P “Finally, she submitted it.” Q “She first wrote a draft.” R “Then she corrected the errors.”

A34.4 Additional practice: Change to passive: “The teacher is checking the papers.”

A34.5 Additional practice: Report: He said, “I can solve it.”

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Preserve tense in passive voice: object + correct form of be + past participle. Reported speech changes depend on context; use conventional backshift when reporting a past statement.

A34.1 “The notebooks were checked by the teacher.”

A34.2 Ravi said that he was tired. Pronoun and tense shift appropriately.

A34.3 Q-R-P. First → then → finally.

A34.4 “The papers are being checked by the teacher.” Present continuous passive uses am/is/are + being + past participle; the auxiliary agrees with the new subject.

A34.5 He said that he could solve it. In ordinary backshift after a past reporting verb, can→could and I→he; preserve the intended speaker reference.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A35 • ORIGINAL PRACTICE

English: comprehension and cloze

Observed: P35 · TGT Computer Science 2021 file 3 (2021) Q96-100

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A35.1 Passage: “A school opened its library for an extra hour. Visits rose, but the librarian said a longer trial was needed before judging reading improvement.” What definitely rose?

A35.2 From the same passage, can we conclude that every visitor read more books?

A35.3 Best title: “Longer Library Hours: Early Results” or “Reading Problems Completely Solved”?

A35.4 Fill: “Practice improves speed, ___ accuracy still requires careful checking.”

A35.5 Additional practice: Passage: “Trees reduce summer heat through shade and evaporation. Removing them may raise local temperatures.” Which inference is supported?

A35.6 Additional practice: Fill the cloze: “Although the task was difficult, she ___ until it was completed.” Choose persisted / surrendered / vanished.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Read the question, locate evidence and distinguish stated fact from inference. Avoid choices that add extreme words such as always or never without support.

A35.1 Library visits. Reading improvement was not yet established.

A35.2 No. Visits do not prove individual reading outcomes.

A35.3 “Longer Library Hours: Early Results.” It reflects the limited evidence.

A35.4 But. The second clause contrasts with the first.

A35.5 Trees can help cool their surroundings. Both stated mechanisms support this; the passage does not prove that trees eliminate every heat problem.

A35.6 Persisted. The contrast introduced by “although” and eventual completion require continued effort.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A36 • ORIGINAL PRACTICE

Hindi: word classes, gender and number

Observed: P35 · TGT Computer Science 2021 file 3 (2021) Q61-62,68

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A36.1 “सीता धीरे चलती है।” में “धीरे” का शब्द-भेद क्या है?

A36.2 “कवि” का मानक स्त्रीलिंग लिखिए।

A36.3 “लड़का” का बहुवचन सामान्य रूप क्या है?

A36.4 Additional practice: “वह प्रतिदिन विद्यालय जाता है।” में “प्रतिदिन” का शब्द-भेद बताइए।

A36.5 Additional practice: “नदी” का बहुवचन लिखिए।

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Questions remain in Hindi because the Hindi test assesses Hindi. Explanations here are in English. Identify the role of a word in its sentence, not only its dictionary label.

A36.1 क्रिया-विशेषण (adverb). It tells how she walks.

A36.2 कवयित्री. Memorise the correct spelling; avoid कवियत्री.

A36.3 लड़के. The oblique plural before a postposition is लड़कों, as in लड़कों को.

A36.4 Adverb (क्रिया-विशेषण), because it specifies how often the going occurs. Identify its role in the sentence rather than its isolated form.

A36.5 नदियाँ in the direct plural; नदियों before a postposition. The form depends on grammatical context, so do not interchange direct and oblique plural forms.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A37 • ORIGINAL PRACTICE

Hindi: कारक, verb and agreement

Observed: P35 · TGT Computer Science 2021 file 3 (2021) Q62,69-70

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A37.1 “राम ने चाकू से फल काटा।” में “चाकू से” कौन-सा कारक है?

A37.2 “पेड़ से पत्ता गिरा।” में “पेड़ से” कौन-सा कारक है?

A37.3 शुद्ध कीजिए: “सीता ने पुस्तक पढ़ा।”

A37.4 Additional practice: “सीमा ने पत्र लिखा।” में “ने” किस कारक का चिह्न है?

A37.5 Additional practice: Correct: “लड़कियाँ खेल रहा है।”

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Look at meaning as well as the marker: “से” can indicate instrument (करण) or separation (अपादान). In sentences with ने, verb agreement needs attention to object marking.

A37.1 करण कारक. The knife is the instrument.

A37.2 अपादान कारक. The leaf separates from the tree.

A37.3 “सीता ने पुस्तक पढ़ी।” The unmarked feminine object पुस्तक controls the participle here.

A37.4 कर्ता कारक. “ने” marks the agent in this perfective transitive construction; पत्र is the object.

A37.5 “लड़कियाँ खेल रही हैं।” The plural feminine subject requires रही and the plural auxiliary हैं. Check gender, number and auxiliary separately.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A38 • ORIGINAL PRACTICE

Hindi: उपसर्ग, प्रत्यय, संधि and समास

Observed: P35 · TGT Computer Science 2021 file 3 (2021) Q63; sandhi/samas included under grammar coverage

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A38.1 “अशुद्ध” में उपसर्ग तथा “मिठास” में प्रत्यय बताइए।

A38.2 “विद्यालय” का संधि-विच्छेद कीजिए।

A38.3 “राजपुत्र” और “माता-पिता” का समास बताइए।

A38.4 Additional practice: “विद्यालय” का संधि-विच्छेद कीजिए।

A38.5 Additional practice: “राजपुत्र” में कौन-सा समास है?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Prefix precedes the root, suffix follows it. Expand a compound before naming its type. In sandhi, identify the original sound boundary.

A38.1 अ-; -आस. अशुद्ध means not pure; मिठास names the quality of sweetness.

A38.2 विद्या + आलय. आ+आ becomes आ (दीर्घ स्वर संधि).

A38.3 राजपुत्र: तत्पुरुष (राजा का पुत्र). माता-पिता: द्वंद्व (माता और पिता).

A38.4 विद्या + आलय. आ + आ becomes आ under दीर्घ स्वर संधि. Reverse the vowel combination rather than splitting at an arbitrary letter.

A38.5 षष्ठी तत्पुरुष: राजा का पुत्र. Expand the compound to identify the suppressed relationship.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A39 • ORIGINAL PRACTICE

Hindi: पर्याय, विलोम, word pairs and substitution

Observed: P35 · TGT Computer Science 2021 file 3 (2021) Q64-67

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A39.1 “जल” का पर्याय और “उदय” का विलोम लिखिए।

A39.2 “दिन” और “दीन” के अर्थ बताइए।

A39.3 “जो कभी न मरे” के लिए एक शब्द लिखिए।

A39.4 Additional practice: “अंधकार” का विलोम और “अग्नि” का एक पर्याय लिखिए।

A39.5 Additional practice: “जो कभी न मरे” के लिए एक शब्द लिखिए।

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Learn meaning in pairs and confirm whether the question asks for the correct or incorrect pair. Similar sounds can hide different meanings.

A39.1 नीर; अस्त.

A39.2 दिन = day; दीन = poor, helpless or humble according to context. Vowel length changes meaning.

A39.3 अमर.

A39.4 प्रकाश; अनल. These are vocabulary recall items; use short recall cards and repeated retrieval rather than a numerical shortcut.

A39.5 अमर. Look for the negative prefix अ- with मर; do not substitute a word meaning merely long-lived.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A40 • ORIGINAL PRACTICE

Hindi: sentence structure, spelling and punctuation

Observed: P35 · TGT Computer Science 2021 file 3 (2021) Q70-73

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A40.1 “जो मेहनत करता है, वह सफल होता है।” सरल, संयुक्त या मिश्र?

A40.2 सही वर्तनी चुनिए: आशीर्वाद / आशिर्वाद।

A40.3 “वाह तुम जीत गए” में उचित विराम-चिह्न लगाइए।

A40.4 Additional practice: Correct the spelling: “आर्शीवाद”.

A40.5 Additional practice: “वह आया और मैं गया।” किस प्रकार का वाक्य है?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

A simple sentence has one main clause; a compound joins independent clauses; a complex sentence contains a subordinate clause. Punctuation should match the sense.

A40.1 मिश्र वाक्य. जो मेहनत करता है is a dependent clause.

A40.2 आशीर्वाद.

A40.3 “वाह! तुम जीत गए।” An exclamation follows the emotional expression.

A40.4 आशीर्वाद. Learn the placement of र् in the standard spelling; sound alone can be misleading.

A40.5 संयुक्त वाक्य. Two independent clauses are joined by और. A मिश्र वाक्य instead contains a dependent clause.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A41 • ORIGINAL PRACTICE

Hindi: idioms, proverbs and comprehension

Observed: P35 · TGT Computer Science 2021 file 3 (2021) Q74-80

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A41.1 “नाक कटना” का अर्थ क्या है?

A41.2 “जैसी करनी वैसी भरनी” का भावार्थ क्या है?

A41.3 गद्यांश: “रीना रोज थोड़ा पढ़ती थी। परीक्षा से पहले उसने अपने कठिन प्रश्न दोहराए।” रीना ने परीक्षा से पहले क्या किया?

A41.4 Additional practice: “नाक कटना” मुहावरे का अर्थ बताइए।

A41.5 Additional practice: गद्यांश: “रीमा ने प्रतिदिन थोड़ा अभ्यास किया। महीने के अंत में उसकी गति और शुद्धता बढ़ गई।” सुधार का मुख्य कारण क्या है?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Recall the figurative sense; do not translate word by word. For an unseen passage, the passage itself is the evidence.

A41.1 अपमान होना: to lose honour or suffer humiliation.

A41.2 One faces the consequences of one’s actions.

A41.3 उसने कठिन प्रश्न दोहराए। The answer is stated directly; do not add that she studied all night.

A41.4 To suffer disgrace or lose honour. Interpret it figuratively in ordinary idiomatic use.

A41.5 Regular practice. The passage directly links daily practice with improved speed and accuracy; it does not mention a new teacher or equipment.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A42 • ORIGINAL PRACTICE

General awareness: history and chronology

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q11-13; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q26-30

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A42.1 Arrange chronologically: Quit India, Dandi March, Non-Cooperation.

A42.2 Who founded the Maurya Empire?

A42.3 Which Harappan site is especially associated with a dockyard in standard school history?

A42.4 Additional practice: Arrange chronologically: Non-Cooperation Movement, Quit India Movement, Civil Disobedience Movement.

A42.5 Additional practice: Match the founder: Brahmo Samaj—Raja Rammohan Roy or Swami Dayanand Saraswati?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Build short event chains and place-person-period maps. Do not memorise an old key blindly: historical question wording and answer keys can contain errors.

A42.1 Non-Cooperation (1920), Dandi March (1930), Quit India (1942). Anchor the sequence 20-30-42.

A42.2 Chandragupta Maurya. Distinguish him from the Gupta ruler Chandragupta I.

A42.3 Lothal, in Gujarat. Link site-state-feature.

A42.4 Non-Cooperation (1920), Civil Disobedience (1930), Quit India (1942). Anchor movements to their starting years; do not order them alphabetically.

A42.5 Raja Rammohan Roy. Dayanand Saraswati founded Arya Samaj. Prepare paired institutions together to avoid exchanging founders.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A43 • ORIGINAL PRACTICE

General awareness: Constitution and polity

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q17-18; verify using Constitution source in guide

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A43.1 Match Articles 14,21 and32 to equality,life/personal liberty and constitutional remedies.

A43.2 Which constitutional amendment introduced Panchayati Raj provisions?

A43.3 Which article contains special provisions relating to Delhi?

A43.4 Additional practice: Which constitutional provision protects life and personal liberty: Article 19 or Article 21?

A43.5 Additional practice: Fundamental Duties appear in which part of the Constitution?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Study article-topic and institution-function pairs. These are memory questions; the shortcut is elimination using the constitutional category, not guessing from a nearby article number.

A43.1 14=equality; 21=life and personal liberty; 32=constitutional remedies.

A43.2 73rd Amendment. Pair 73 with rural Panchayats and 74 with urban municipalities.

A43.3 Article 239AA. Do not confuse it with Article 239 (Union territories generally).

A43.4 Article 21. Keep compact article-purpose pairs; Article 19 addresses specified freedoms rather than the wording asked here.

A43.5 Part IVA, principally Article 51A. Link the part with the article to solve both styles of recall question.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A44 • ORIGINAL PRACTICE

General awareness: geography and environment

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q8-10; P35 · TGT Computer Science 2021 file 3 (2021) Q8-10

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A44.1 Which major peninsular rivers flow west into the Arabian Sea: Narmada, Tapi, Godavari?

A44.2 Which soil is strongly associated with cotton cultivation in the Deccan?

A44.3 What is the difference between in-situ and ex-situ conservation?

A44.4 Additional practice: Which of Narmada and Godavari drains into the Arabian Sea?

A44.5 Additional practice: Which atmospheric layer contains most weather phenomena?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Use maps: river-source-direction-mouth, crop-climate and soil-crop. Separate weather from long-term climate. Link conservation terms to what they protect.

A44.1 Narmada and Tapi. Godavari flows east to the Bay of Bengal.

A44.2 Black soil (regur), with high moisture-retention capacity.

A44.3 In-situ protects organisms in their habitat; ex-situ protects them outside it, such as seed banks.

A44.4 Narmada. Godavari drains into the Bay of Bengal. Trace origin→direction→mouth on a map rather than memorising names alone.

A44.5 Troposphere. Temperature generally falls with altitude within it; most atmospheric water vapour is concentrated in the lower atmosphere.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A45 • ORIGINAL PRACTICE

General awareness: everyday science

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q5-7; P35 · TGT Computer Science 2021 file 3 (2021) Q5-7

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A45.1 Which blood cells mainly defend against pathogens?

A45.2 Why do stars twinkle?

A45.3 An acidic aqueous solution has pH below or above 7 at ordinary school-level conditions?

A45.4 Additional practice: A person with blood group AB has which ABO antigens on red blood cells?

A45.5 Additional practice: Which gas is usually released when an active metal reacts with a dilute non-oxidising acid?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Learn cause-effect pairs across biology, chemistry and physics. Units and mechanisms eliminate plausible-sounding distractors.

A45.1 White blood cells. Red cells mainly transport oxygen; platelets help clotting.

A45.2 Atmospheric refraction through changing air layers. It is not due to stars rapidly switching off.

A45.3 Below 7. Neutral water is approximately pH 7 at 25°C.

A45.4 Both A and B antigens. Keep antigens on cells separate from antibodies in plasma; compatibility questions also depend on the blood component involved.

A45.5 Hydrogen. The metal displaces hydrogen from the acid; this rule is not universal for every metal or oxidising acid.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A46 • ORIGINAL PRACTICE

General awareness: economy and organisations

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q3-4,15-16; P35 · TGT Computer Science 2021 file 3 (2021) Q3-4

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A46.1 Which organisation publishes the World Economic Outlook?

A46.2 Are income tax and GST both direct taxes?

A46.3 What does GDP measure in the standard production definition?

A46.4 Additional practice: Which organisation publishes the World Economic Outlook?

A46.5 Additional practice: If nominal GDP rises by 10% and the price index rises by 10%, what is the approximate real GDP change?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Separate institution-function, tax-type and report-publisher. For current rates/ranks, always keep the reference date; an older exam answer may no longer be current.

A46.1 International Monetary Fund (IMF). World Development Report is associated with the World Bank.

A46.2 No. Income tax is direct; GST is indirect.

A46.3 Value of final goods and services produced within a country’s domestic territory during a specified period. “Final” avoids double counting.

A46.4 International Monetary Fund (IMF). Pair report with publisher; do not replace it with the World Bank's Global Economic Prospects.

A46.5 Zero; exactly zero when both factors are 1.10. Real growth=(1.10/1.10−1)×100. Subtracting rates is only an approximation in general.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A47 • ORIGINAL PRACTICE

General awareness: culture, sports and institutions

Observed: P33 · TGT Computer Science 2021 file 1 (2021) Q1-2,19-20; P09 · TGT MATHS FEMALE 22nd September 2018 Shift 1 (2018) Q31-34

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A47.1 Match Bharatanatyam and Kathakali to their principal states of origin.

A47.2 Flute, tabla and sitar belong to which broad instrument families?

A47.3 Match Wimbledon and the Thomas Cup to sports.

A47.4 Additional practice: Flute, tabla and sitar: which is a wind instrument?

A47.5 Additional practice: The Thomas Cup is associated with which sport?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Use compact maps: dance-state, instrument-family, monument-location, sport-trophy, organisation-HQ. No mathematical shortcut substitutes for recall.

A47.1 Tamil Nadu; Kerala, respectively.

A47.2 Wind, percussion/membrane, string, respectively.

A47.3 Wimbledon=tennis; Thomas Cup=men’s badminton team competition.

A47.4 Flute, a sushira vadya. Tabla is a percussion instrument and sitar is a string instrument. Classify by how sound is produced.

A47.5 Badminton, specifically the men's international team competition. Prepare the related Uber Cup for the women's team competition.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A48 • ORIGINAL PRACTICE

Partnership, variation and replacement mixtures

Ratio/time-work syllabus extensions; replacement analogue P20 · TGT SOCIAL SCIENCE FEMALE 15th Oct 2018 Shift 1 (2018) Q51

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A48.1 A invests Rs 5,000 for 12 months; B Rs 10,000 for 3 months. Split Rs 9,000 profit.

A48.2 A 20-litre container of milk has 5 litres removed and replaced with water twice. Milk left?

A48.3 Six identical machines need 10 hours. How long for 15 machines at the same rate?

A48.4 Additional practice: A invests ₹8,000 for 12 months and B ₹12,000 for 8 months. Divide ₹10,000 profit.

A48.5 Additional practice: A container holds 40 litres of pure milk. Remove 10 litres and replace with water, twice. How much milk remains?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Partnership profit normally follows capital×time. Direct variation means y/x constant; inverse variation means xy constant. Repeated replacement leaves original fraction (1-v/V)^n after n equal replacements.

A48.1 Ratio=60000:30000=2:1. A=Rs 6,000,B=Rs 3,000.

A48.2 11.25 litres. $20\times(1-5/20)^2=20\times9/16$.

A48.3 Four hours. Inverse variation: $6\times10/15$.

A48.4 ₹5,000 each. Capital×time products are equal: 96,000 and 96,000. Use money-time contributions if investment periods differ.

A48.5 22.5 litres. Remaining fraction each time=$1-10/40=3/4$; milk=$40(3/4)^2$. Assume thorough mixing before the second removal.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A49 • ORIGINAL PRACTICE

Algebraic identities, surds and proportion

Observed: P09 · TGT MATHS FEMALE 22nd September 2018 Shift 1 (2018) Q45; P20 · TGT SOCIAL SCIENCE FEMALE 15th Oct 2018 Shift 1 (2018) Q42

QUESTIONS FIRST · PATTERN-BASED PRACTICE

A49.1 If $x+1/x=5$,find x^2+1/x^2.

A49.2 Rationalise $1/(\sqrt{5}+2)$.

A49.3 Find the mean proportional between 4 and 9.

A49.4 Additional practice: If $x+1/x=5$, find x^2+1/x^2.

A49.5 Additional practice: Simplify $1/(\sqrt{5}-2)$.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Use $(a+b)^2=a^2+2ab+b^2$; $a^2-b^2=(a-b)(a+b)$. Rationalise by the conjugate. Mean proportional between positive a,b is $\sqrt{ab}$.

A49.1 23. Square and subtract 2.

A49.2 $\sqrt{5}$-2. Multiply numerator and denominator by $\sqrt{5}$-2; denominator=5-4=1.

A49.3 6. $\sqrt{(4\times9)}$. The third proportional is different: 4:9=9:x gives x=81/4.

A49.4 23. Square: $25=x^2+2+1/x^2$; subtract 2. The nonzero-x condition is implicit in the original expression.

A49.5 $\sqrt{5}$+2. Multiply numerator and denominator by the conjugate; denominator=5−4=1.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

A50 • ORIGINAL PRACTICE

Visual reasoning: original figure drill

Original visual extension; observed figure formats P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q13-15 and P13 · TGT NATURAL SCIENCE FEMALE 27th September 2018 Shift 1 (2018) Q4

QUESTIONS FIRST · PATTERN-BASED PRACTICE

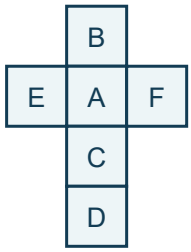


Figure 1: cube net


Figure 2: arrow and mirror

A50.1 In Figure 1, the net folds into a cube. Which face is opposite A?

A50.2 In Figure 2, the arrow points to the upper-right. A vertical mirror is placed to its right. Which direction will the reflected arrow point?

A50.3 A square is folded in half vertically and then horizontally. A hole is punched strictly inside the folded square, away from folds. How many holes after unfolding?

A50.4 Additional practice: A cube net has vertical squares A-B-C-D and squares E and F attached left and right of B. Which face is opposite B?

A50.5 Additional practice: A transparent sheet has a dot 2 cm to the left of a vertical fold line. Where does its mirror image lie?

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

Use the diagrams printed with this unit. Mentally fold one square at a time; opposite faces never share an edge. In a vertical mirror, x-coordinates reverse while heights stay unchanged.

A50.1 D. B and C fold to opposite sides; E and F form the other opposite pair; D closes opposite A.

A50.2 Upper-left. Left-right reverses but up-down does not.

A50.3 Four. Each unfolding doubles the hole count; points on a fold would need separate treatment.

A50.4 D. Fold the strip around the cube: A opposes C, B opposes D, and E opposes F. Verify the actual net arrangement before memorising an opposite-face rule.

A50.5 2 cm to the right of the line, at the same height. Reflection preserves perpendicular distance to the mirror line.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

Verified PYQ bridge: arithmetic and reasoning

Official paper references P33 · TGT Computer Science 2021 file 1 (2021) Q45; P35 · TGT Computer Science 2021 file 3 (2021) Q49; P28 · Question Paper for Post: TGT Computer Science, Post Code: 39/22, Date: 24.06.2023, Shift-2 (2023) Q2; P09 · TGT MATHS FEMALE 22nd September 2018 Shift 1 (2018) Q16

QUESTIONS FIRST · PATTERN-BASED PRACTICE

AP1.1 P33 Q45: Two numbers multiply to 1690; HCF is two-fifths of LCM. Find HCF.

AP1.2 P35 Q49: Sale at Rs 864 gives a 28% loss. What sale price gives 20% profit?

AP1.3 P28 Q2: In a 50-student row, Ravi is 20th from right; Rahul is 15 places left of him. Rahul's left rank?

AP1.4 P09 Q16: Four years ago two ages were 2:1; five years later they will be 3:2. Find the younger person's present age.

SOLUTIONS · SHORTCUT METHOD · WHY IT WORKS

The following are shortened paraphrases of identifiable past-paper items, not original practice. Attempt the linked official item to see its full options and bilingual wording.

AP1.1 26. Let H=h,L=5h/2. Then (5/2)h²=1690, so h²=676. Check L=65.

AP1.2 Rs 1,440. CP=864/0.72=1200; desired SP=1200×1.2.

AP1.3 16th. Ravi=50-20+1=31; Rahul=31-15.

AP1.4 13 years. Let younger now=x,older now=y. y-4=2(x-4), and 2(y+5)=3(x+5). Substitute y=2x-4 to get x=13.

Review: □ attempted □ correct without help □ redo tomorrow □ redo in 3 days □ redo in 7 days

Sources and further checks

Paper IDs refer to the official files in the 30-day plan’s source register. A question number is the number printed in that file, not a coaching-site serial number. Old answer-key colours may not survive text copying. Open the PDF to inspect the original.

P02 · 46 TGT HINDI FEMALE 28th Aug 2018 Shift1.pdf Download

P09 · 57 TGT MATHS FEMALE 22nd September 2018 Shift 1.pdf Download

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